

Δ vs frozen (good-oriented, value + 95% CI) · PRETRAIN block / SURGERY block · SCORE = mean per-metric-normalized (0-1) · BLUE BOX + big Δ = best arm per metric
WINNER row (champion duel — S=surgery, P=pretrain, ==tie): surgery 5 · pretrain 2 · tie 2 → SURGERY

ViT-g pretrain_2X encoder	+5.81 [+3.671, +8.055]	+0.037 [+0.036, +0.039]	-0.029 [-0.044, -0.014]	-0.0022 [-0.003, -0.001]	+0.0023 [+0.002, +0.002]	-0.0016 [-0.002, -0.001]	-0.0011 [-0.001, -0.001]	+0.0054 [+0.005, +0.006]	+0.0015 [+0.001, +0.002]	0.49
ViT-g pretrain encoder	+4.00 [+1.699, +6.356]	+0.035 [+0.034, +0.036]	-0.021 [-0.035, -0.006]	+0.0046 [+0.004, +0.005]	+0.0019 [+0.002, +0.002]	+0.0035 [+0.003, +0.004]	-0.0009 [-0.001, -0.001]	+0.0057 [+0.005, +0.006]	-0.0001 [-0.000, +0.000]	0.60
ViT-g pretrain head	+0.329 [-1.753, +2.411]	+0.0044 [+0.004, +0.005]	-0.012 [-0.025, +0.001]	-0.0002 [-0.001, +0.000]	0 [+nan, +nan]	0 [+nan, +nan]	0 [+nan, +nan]	0 [+nan, +nan]	0 [+nan, +nan]	0.30
ViT-g surgery 3stage_DI encoder	+5.37 [+3.233, +7.562]	+0.037 [+0.035, +0.038]	-0.026 [-0.041, -0.010]	+0.0082 [+0.007, +0.009]	+0.0014 [+0.001, +0.002]	+0.0047 [+0.004, +0.005]	-0.0016 [-0.002, -0.001]	+0.0037 [+0.003, +0.004]	-0.0009 [-0.001, -0.001]	0.54
ViT-g surgery 3stage_DI head	+4.05 [+1.918, +6.247]	+0.043 [+0.042, +0.045]	-0.020 [-0.035, -0.005]	+0.0044 [+0.004, +0.005]	+0.0019 [+0.002, +0.002]	+0.0035 [+0.003, +0.004]	-0.0009 [-0.001, -0.001]	+0.0057 [+0.005, +0.006]	-0.0001 [-0.000, +0.000]	0.62
ViT-g surgery noDI encoder	+4.22 [+1.973, +6.630]	+0.050 [+0.048, +0.052]	-0.025 [-0.040, -0.009]	+0.0007 [-0.000, +0.002]	+0.0022 [+0.002, +0.002]	-0.0024 [-0.003, -0.002]	-0.0012 [-0.001, -0.001]	+0.0072 [+0.007, +0.008]	+0.0034 [+0.003, +0.004]	0.59
ViT-g surgery noDI head	+4.71 [+2.356, +7.069]	+0.056 [+0.054, +0.058]	-0.023 [-0.038, -0.008]	+0.0042 [+0.003, +0.005]	+0.0019 [+0.002, +0.002]	+0.0035 [+0.003, +0.004]	-0.0009 [-0.001, -0.001]	+0.0057 [+0.005, +0.006]	-0.0001 [-0.000, +0.000]	0.65
WINNER	P	S	P	S	=	S	=	S	S	
	action_top1 ↑ better classify motion class from frozen features	motion_cos ↑ better same-motion clips cluster vs different motions	taxonomy_f1 ↑ better read scene attributes from features	future_rmse ↓ better predictor guesses masked next-frame tokens	rollout ↓ better error growth over multi-step rollout	causal ↓ better predict 2nd half from 1st half	tdist ↓ better accuracy decay vs prediction horizon	teacher_free ↓ better free-run minus teacher-forced gap	maskratio ↓ better degradation under heavier masking	SCORE (norm 0-1)